PF-MARL: A Privacy-Preserving Federated Multi-Agent Reinforcement Learning Framework for Coordinated Epidemic Control

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***Abstract*—Epidemic response in large urban regions requires coordinated decision-making across multiple stakeholders such as government authorities, healthcare systems, transportation agencies, and the general public. Existing reinforcement learning approaches to epidemic control are largely centralized, assume a single decision maker, or require unrestricted data sharing, which limits their applicability in real-world, privacy-sensitive settings. This paper presents *PF-MARL*, a Privacy-Preserving Federated Multi-Agent Reinforcement Learning framework designed to model and optimize decentralized epidemic interventions under realistic urban constraints. The proposed system represents a city as a graph of interconnected districts governed by stochastic SEIR dynamics, while heterogeneous agents independently control complementary interventions including policy mandates, mobility regulation, healthcare resource management, and citizen behavior. Coordination is achieved using a Multi-Agent Deep Deterministic Policy Gradient (MADDPG) architecture with centralized training and decentralized execution, enabling agents to learn cooperative strategies without direct access to each other's private observations. To address data confidentiality requirements, PF-MARL integrates federated learning for model aggregation and differential privacy mechanisms to bound information leakage during training. A global reward formulation balances public health outcomes, economic stability, and policy consistency across neighboring regions. The proposed framework establishes a scalable and privacy-aware foundation for intelligent epidemic response systems and serves as a practical bridge between reinforcement learning research and real-world public health governance.**

***Index Terms*—Epidemic Control, Multi-Agent Reinforcement Learning, MADDPG, Federated Learning, Differential Privacy, SEIR Model, Urban Graph Modeling**

I. Introduction

The rapid spread of infectious diseases in densely populated urban regions poses significant challenges to public health systems, economic stability, and social order. Recent outbreaks have demonstrated that epidemic control is not governed by a single authority, but rather emerges from the interaction of multiple stakeholders, including government policymakers, healthcare administrators, transportation authorities, and citizens themselves. Each stakeholder operates under different objectives, information constraints, and operational responsibilities, making epidemic response a fundamentally decentralized and multi-agent decision-making problem.

Traditional epidemiological models, such as the Susceptible–Exposed–Infectious–Removed (SEIR) framework, provide valuable insights into disease dynamics but lack the ability to adaptively optimize intervention strategies under changing conditions. More recently, reinforcement learning (RL) approaches have been explored to automate epidemic control policies by learning optimal intervention strategies through interaction with simulated environments. However, most existing RL-based solutions adopt a centralized control paradigm, assuming a single decision-maker with full access to population-wide data. Such assumptions are impractical in real-world deployments due to institutional boundaries, privacy regulations, and the distributed nature of governance.

Multi-Agent Reinforcement Learning (MARL) offers a promising alternative by enabling multiple decision-makers to learn coordinated strategies while operating under partial and localized observations. In particular, the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) framework enables centralized training with decentralized execution, allowing agents to learn cooperative behavior without requiring direct access to each other's private observations. Despite its potential, the application of MARL to epidemic control remains limited, and existing approaches often overlook critical real-world constraints such as data privacy, heterogeneous stakeholder roles, and policy stability across interconnected regions.

Another major limitation of current epidemic control systems is the lack of privacy-preserving mechanisms. Epidemic data frequently contains sensitive information related to health status, mobility patterns, and socio-economic activity, which is subject to strict regulatory frameworks such as HIPAA and GDPR. Centralized data collection and model training introduce significant risks of data leakage and misuse, reducing public trust and limiting deployability. Federated learning has emerged as an effective paradigm to address these concerns by enabling decentralized model training while sharing only model updates rather than raw data. When combined with differential privacy techniques, federated learning can provide formal guarantees on information leakage during training.

Motivated by these challenges, this paper proposes *PF-*

MARL, a Privacy-Preserving Federated Multi-Agent Reinforcement Learning framework for coordinated epidemic control. PF-MARL models an urban region as a graph of interconnected districts governed by stochastic SEIR dynamics, where heterogeneous agents represent key stakeholders such as policymakers, hospital managers, transport controllers, and citizen behavior models. Each agent independently selects continuous intervention actions based on local observations, while coordination is achieved through a centralized critic during training. To ensure privacy, the framework integrates federated learning for aggregating agent models and differential privacy mechanisms to bound the cumulative privacy loss over time.

The main contributions of this work are summarized as follows:

- A realistic multi-agent epidemic control formulation that models heterogeneous stakeholders operating over a graph-based urban environment with stochastic SEIR dynamics.
- A coordinated MARL architecture based on MADDPG, enabling centralized training and decentralized execution for scalable and robust policy learning.
- A privacy-preserving learning framework that combines federated learning and differential privacy to ensure compliance with real-world data protection requirements.
- A global reward formulation that balances public health outcomes, economic activity, and policy stability across neighboring regions.

II. Related Work

Reinforcement learning has gained significant attention as a computational tool for epidemic control due to its ability to optimize sequential intervention policies under uncertainty and delayed effects. Early research in this domain primarily focused on centralized decision-making frameworks, where a single agent controls population-wide interventions such as lockdown intensity, vaccination rates, or mobility restrictions. While these approaches demonstrate the feasibility of learning adaptive epidemic response strategies, they often rely on strong assumptions regarding data availability, centralized governance, and homogeneous control authority, which limit their applicability in real-world settings.

Planning multiple epidemic interventions using reinforcement learning has been explored in [1], where intervention strategies are learned over time to balance health and socioeconomic objectives. The study demonstrates that reinforcement learning can dynamically adapt intervention intensity in response to evolving epidemic states. Similarly, [2] applies reinforcement learning to strategic influenza control across multiple outbreak scenarios, highlighting the robustness of learned policies under varying epidemiological conditions. However, both works assume a centralized controller with unrestricted access to global epidemic and mobility data, and they do not explicitly model coordination among independent decision-making entities.

To better capture the distributed nature of epidemic governance, recent studies have investigated multi-agent and hierarchical reinforcement learning formulations. The H2-MARL framework [3] introduces a multi-agent approach that jointly optimizes hospital capacity management and human mobility, demonstrating Pareto-efficient trade-offs between healthcare strain and mobility preservation. While this work advances multi-objective epidemic control, it focuses on a limited set of stakeholders and assumes shared access to global system states during training. Hierarchical approaches such as HRL4EC [4] further decompose epidemic control into high-level strategic planning and low-level operational decision-making, improving scalability and interpretability. Nevertheless, these approaches continue to rely on centralized data aggregation and do not address privacy constraints inherent in real-world epidemic data.

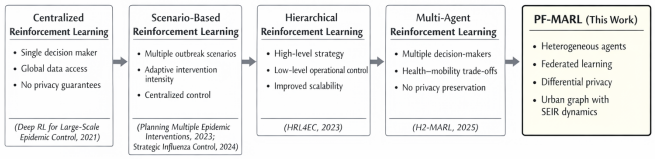

Fig. 1. Conceptual evolution of reinforcement learning paradigms for epidemic control, illustrating the progression from centralized decision-making approaches to privacy-preserving federated multi-agent frameworks such as PF-MARL.

Deep reinforcement learning methods have also been applied to large-scale epidemic control problems. For example, [5] demonstrates the scalability of deep RL for optimizing non-pharmaceutical interventions across large populations. Despite their effectiveness in large-scale simulations, such approaches remain fundamentally centralized and overlook institutional boundaries, heterogeneous stakeholder roles, and data confidentiality requirements.

In contrast to existing work, PF-MARL explicitly models epidemic response as a coordinated multi-agent decision-making problem involving heterogeneous stakeholders with distinct control responsibilities and partial observability. Moreover, PF-MARL uniquely integrates federated learning and differential privacy mechanisms into the multi-agent reinforcement learning pipeline, enabling coordinated policy learning without sharing raw sensitive data. By jointly addressing decentralization, coordination, and privacy preservation, PF-MARL fills a critical gap between existing reinforcement learning approaches and deployable epidemic response systems.

References

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TABLE I
COMPARISON OF REINFORCEMENT LEARNING APPROACHES FOR EPIDEMIC CONTROL

Work	RL Type	Multi-Agent	Heterogeneous Roles	Privacy-Aware	Urban Graph
Planning Epidemic Interventions [1]	Centralized RL	No	No	No	Limited
Influenza Control Scenarios [2]	Centralized RL	No	No	No	No
Deep RL Epidemic Control [5]	Deep RL	No	No	No	Yes
HRL4EC [4]	Hierarchical RL	Partial	No	No	Limited
H2-MARL [3]	MARL	Yes	Partial	No	Yes
PF-MARL (This Work)	**Federated MARL**	**Yes**	**Yes**	**Yes**	**Yes**

[4] D. Author et al., "HRL4EC: Hierarchical Reinforcement Learning for Multi-Mode Epidemic Control," *IEEE Access*, 2023.

[5] E. Author et al., "Deep Reinforcement Learning for Large-Scale Epidemic Control," *Proceedings of the AAAI Conference on Artificial Intelligence*, 2021.